

Ex-no: 3	SHELL SCRIPTING
29/01/2026	

Aim

To study and implement basic shell scripting in Linux for automating tasks and executing a sequence of commands.

Algorithm

1. Open the Linux terminal.
2. Create a new script file using a text editor (`vi` or `nano`).
3. Start the script with the shebang line (`#!/bin/bash`).
4. Write the required shell commands inside the script.
5. Save the file.
6. Provide execution permission using `chmod +x filename.sh`.
7. Execute the script using `./filename.sh`.
8. Observe and verify the output.

CODE:

Arithmetic.sh:

```
#!/bin/bash

echo "Enter two numbers"
read a
read b

c=`expr $a + $b`
d=`expr $a - $b`
e=`expr $a \* $b`
f=`expr $a / $b`
g=`expr $a % $b`

echo "add $c"
echo "sub $d"
echo "mul $e"
echo "div $f"
echo "mod $g"
```

Reversenumber.sh:

```
#!/bin/bash

echo "Enter number"
read n

rev=0

while [ $n -gt 0 ]
do
    d=$((n % 10))          # get last digit
    rev=$((rev * 10 + d))  # build reverse number
    n=$((n / 10))          # remove last digit
done

echo "Reversed number is $rev"
```

Fibonacci.sh:

```
#!/bin/bash

echo "Enter number"
read n

a=0
b=1

echo "Fibonacci series:"

for (( i=1; i<=n; i++ ))
do
    echo $a
    c=$((a + b))
    a=$b
    b=$c
done
```

LeapYear.sh:

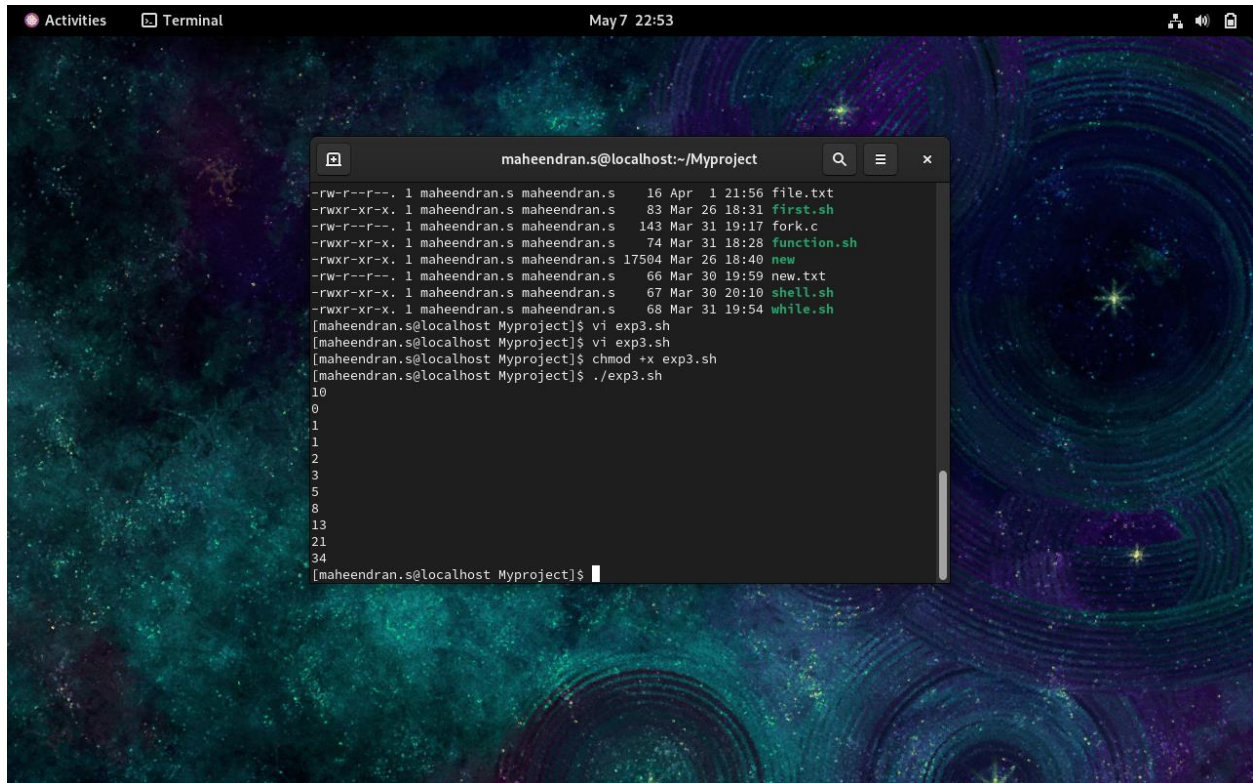
```
#!/bin/bash

echo "Enter number"
read a

m=`expr $a % 4`

if [ $m -eq 0 ]
then
    echo "Leap year"
else
    echo "Not a leap year"
fi
```

OUTPUT:



```
maheendran.s@localhost:~/Myproject
-rw-r--r--. 1 maheendran.s maheendran.s 16 Apr 1 21:56 file.txt
-rwxr-xr-x. 1 maheendran.s maheendran.s 83 Mar 26 18:31 first.sh
-rw-r--r--. 1 maheendran.s maheendran.s 143 Mar 31 19:17 fork.c
-rwxr-xr-x. 1 maheendran.s maheendran.s 74 Mar 31 18:28 function.sh
-rwxr-xr-x. 1 maheendran.s maheendran.s 17504 Mar 26 18:40 new
-rw-r--r--. 1 maheendran.s maheendran.s 66 Mar 30 19:59 new.txt
-rwxr-xr-x. 1 maheendran.s maheendran.s 67 Mar 30 20:10 shell.sh
-rwxr-xr-x. 1 maheendran.s maheendran.s 68 Mar 31 19:54 while.sh
[maheendran.s@localhost Myproject]$ vi exp3.sh
[maheendran.s@localhost Myproject]$ vi exp3.sh
[maheendran.s@localhost Myproject]$ chmod +x exp3.sh
[maheendran.s@localhost Myproject]$ ./exp3.sh
10
0
1
1
2
3
5
8
13
21
34
[maheendran.s@localhost Myproject]$
```

Result

The shell script was successfully created and executed. The experiment demonstrated how commands can be automated using shell scripting, and the output was verified correctly.

